SEMINAR REPORT

A seminar Report on

Industrial Immersive Training Platform

Submitted in partial fulfillment for the award of

THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

By

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FOR THE ACADEMIC YEAR: 2026. – 2027

DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code: 0059)

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(An Affiliated Institute of Maharashtra State Board of Technical Education)

DEPARTMET OF COMPUTER ENGINEERING

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Abstract

Immersive technology is becoming increasingly useful in education, industrial training, and technical skill development. Traditional industrial training may require access to physical machines, trained instructors, and dedicated workshop facilities. Students may not always get sufficient opportunities to understand different industrial machines and their components through practical sessions alone.

The proposed project, “Industrial Immersive Training Platform,” aims to develop a web-based application that helps students, teachers, and factory employees learn about industrial machines through interactive digital content. The platform will provide machine search facilities, machine information, interactive 3D visualization, safety instructions, and simulated troubleshooting exercises.

The proposed system will also include user authentication, machine categories, institution-specific machine management, quizzes, and training progress tracking. It will use web development technologies and 3D visualization tools to make learning accessible through a web browser without requiring a dedicated Virtual Reality (VR) headset.

The main objective of this project is to provide an interactive and accessible learning environment that supports basic industrial machine training before users work with actual equipment. The platform is intended for educational purposes and will not replace practical workshop training or official industrial safety procedures.

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Chapter 1
Introduction

Industrial training is an important part of technical education. It helps students understand industrial machines, their components, working principles, and safety procedures. However, access to real industrial machines may be limited due to their high cost, maintenance requirements, and safety risks.

Immersive technologies, such as Virtual Reality (VR), Augmented Reality (AR), and interactive 3D visualization, can provide new ways to learn about industrial equipment. These technologies allow learners to explore digital machine models and understand their components in an interactive environment.

The proposed project, “Industrial Immersive Training Platform,” aims to develop a web-based application for interactive industrial machine learning. It will include machine information, 3D models, component identification, simulated troubleshooting exercises, safety quizzes, and progress tracking.

The main objective of this project is to make industrial machine learning more accessible and interactive for students and trainees. The platform will support practical education but will not replace actual workshop training or official safety procedures.

Chapter 2
Literature Review

A literature review studies existing research related to a project. This chapter reviews twelve research papers on Virtual Reality (VR), Augmented Reality (AR), immersive technologies, and interactive 3D training systems used in industrial training and maintenance. These papers explain the methods used, their findings, and their limitations.

1. Towards the Adoption of Virtual Reality Training Systems for the Self-Tuition of Industrial Robot Operators: A Case Study at KUKA (2021)

Authors: Filippo Gabriele Pratticò and Fabrizio Lamberti.

Pratticò and Lamberti (2021) developed and evaluated a Virtual Reality Training System for teaching industrial robot maintenance. The system allowed trainees to learn a maintenance procedure in a virtual environment before performing the task on a real robot. The researchers compared VR-based training with traditional classroom and practical training. The results showed that VR training could help learners complete the maintenance task with results comparable to traditional training. However, virtual training still needs improvement in simulating physical tools and human interaction. This study demonstrates the potential of VR for industrial training**.**

2. Methodology for the Field Evaluation of the Impact of Augmented Reality Tools for Maintenance Workers in the Aeronautic Industry (2021)

Authors: Quentin Loizeau et al.

This research focuses on evaluating Augmented Reality tools used by aircraft maintenance workers. AR can display digital instructions and visual guidance over real equipment, helping workers understand maintenance procedures. The study presents a methodology for evaluating the impact of AR tools in an industrial environment. Such tools can support workers by providing instructions directly where they are

needed. However, their effectiveness depends on the design of the instructions, the equipment, and the working environment. This research is relevant to developing visual guidance for industrial machine training.

3. Situated Visualization Towards Manufacturing Maintenance Training: Scoping Review, Design and User Study (2025)

Authors: Zeinab BagheriFard et al.

This study explores the use of situated visualization for manufacturing maintenance training. It examines how 2D labels and 3D visual indicators can help learners understand machine components and maintenance instructions. The researchers reviewed existing work and explored a visualization design through a user study. The approach can make technical information easier to understand by connecting instructions with machine components. However, the usefulness of visualization depends on its design and how clearly information is presented. This study supports the use of labelled 3D machine models in the proposed training platform.

4. Applications of Immersive Technologies for Occupational Safety and Health Training and Education: A Systematic Review (2023)

Authors: Akinloluwa Babalola et al.

This systematic review examines the use of immersive technologies in occupational safety and health training. The researchers reviewed 67 articles related to technologies such as Virtual Reality and Augmented Reality. These technologies allow learners to understand workplace hazards and practise safety procedures in simulated environments. The review highlights their potential to improve safety awareness and provide practical learning experiences. However, the effectiveness of immersive training depends on the quality of the simulation and the learning objectives. This research supports including safety instructions and quizzes in the proposed platform.

5. XR and Workers' Safety in High-Risk Industries: A Comprehensive Review (2025)

Authors: Joana Eva Dodoo et al.

This paper reviews the use of Extended Reality (XR), including VR and AR, to improve worker safety in high-risk industries. These technologies can simulate hazardous situations and help workers understand safety procedures without directly exposing them to workplace risks during practice. The review discusses the potential of XR for safety training and hazard awareness. However, virtual simulations cannot represent every real-world condition and should not replace practical training or official safety procedures. This study supports the inclusion of safety guidance in the Industrial Immersive Training Platform.

6. VR-ManuSafe: Virtual Reality Learning Environment for Manufacturing Safety Training (2026)

Authors: Lei Fengjin, Dalbir Singh, and Lam Meng Chun.

This research presents VR-ManuSafe, a Virtual Reality learning environment designed for manufacturing safety training. It uses a virtual factory environment to help learners understand industrial hazards and different types of workplace injuries. The system provides a simulated setting for learning about manufacturing safety. Virtual environments can help learners explore safety situations without interacting directly with hazardous machinery. However, specialised VR equipment may be required, depending on the implementation. This paper is relevant to developing a virtual industrial environment with safety information and interactive learning features.

7. An Emerging Immersive Technology – A Survey (2020)

Authors: A. Pavithra et al.

This paper surveys emerging immersive technologies and their applications. It discusses how technologies such as Virtual Reality, Augmented Reality, and Mixed Reality can create interactive digital experiences. These technologies are useful in areas such as education, training, simulation, and industrial applications. The study provides a general understanding of immersive technology and its potential benefits. However,

a broad survey does not necessarily provide detailed experimental results for a specific industrial training system. This paper provides background knowledge for selecting suitable immersive technologies for the proposed project.

8. Immersive Technology – Uses, Challenges and Opportunities

Authors: Mandeep Handa, Er. Gagandeep Aul, and Shelja Bajaj.

This paper discusses the uses, challenges, and opportunities associated with immersive technology. It explains how immersive environments can make learning more interactive by allowing users to explore digital content. In industrial training, such technologies can help learners understand machines, components, and operating procedures. The paper also highlights challenges associated with adopting immersive technologies, including technical requirements and implementation difficulties. These challenges are important when designing a training platform that should be accessible and easy to use. The study provides general background for the proposed system.

9. The State of Immersive Technology Research: A Literature Analysis (2018)

Authors: Ayoung Suh and Jane Prophet.

This study analyses existing research on immersive technologies and examines major themes in the field. It explores how immersive technologies are used to create interactive experiences and support different applications. The research helps explain the development and growing importance of immersive systems. Its findings provide a foundation for understanding how these technologies can be applied in education and industry. However, the study focuses broadly on immersive technology research rather than one specific industrial training platform. It helps establish the theoretical background for the proposed project.

10. A Web-Based Digital Twin Framework for Interactive E-Learning in Engineering Education (2025)

Authors: P. Weis, R. Bašťovanský, and M. Vereš.

This research presents a web-based digital twin framework for interactive engineering education. A digital twin is a digital representation of a real-world object or system. The framework allows learners to interact with a digital model through a web-based learning environment. The paper also reports a pilot study involving seven professionals. A web-based approach can make interactive learning more accessible without requiring every learner to install specialised desktop software. However, the pilot involved a small number of participants, so further evaluation is needed. This research is closely related to the proposed browser-based industrial training platform.

11. Evaluation of an Immersive Virtual Learning Environment for Operator Training in Mechanical and Plant Engineering Using Video Analysis (2020)

Authors: C. Pletz and B. Zinn.

This study evaluates an immersive Virtual Reality learning environment designed for operator training in mechanical and plant engineering. The researchers investigated how learners performed training tasks in a virtual environment and assessed their learning process using video analysis. The study involved 13 participants. Immersive training can help learners understand procedures and practise tasks in a simulated environment before working with real equipment. However, the effectiveness of the training depends on the learning task and the ability to transfer knowledge to real-world situations. This study supports using interactive simulations for operator training.

12. Industrial Machine Identification Using Augmented Reality (2021)

Authors: Kavya Shakthi R. P., Kavin Raja A. S., Janani S. R., and Sangeetha K.

This paper presents an approach for identifying industrial machines using Augmented Reality. The system uses technologies such as Unity, Vuforia, and ARCore to support AR-based machine identification. Augmented Reality can display digital information about a machine, helping users recognise equipment and understand its features. Such an approach can make industrial information more accessible and interactive. However, AR-based identification may depend on device compatibility and the accuracy of the identification process. This study is relevant to developing visual machine identification and component information features for the proposed platform.

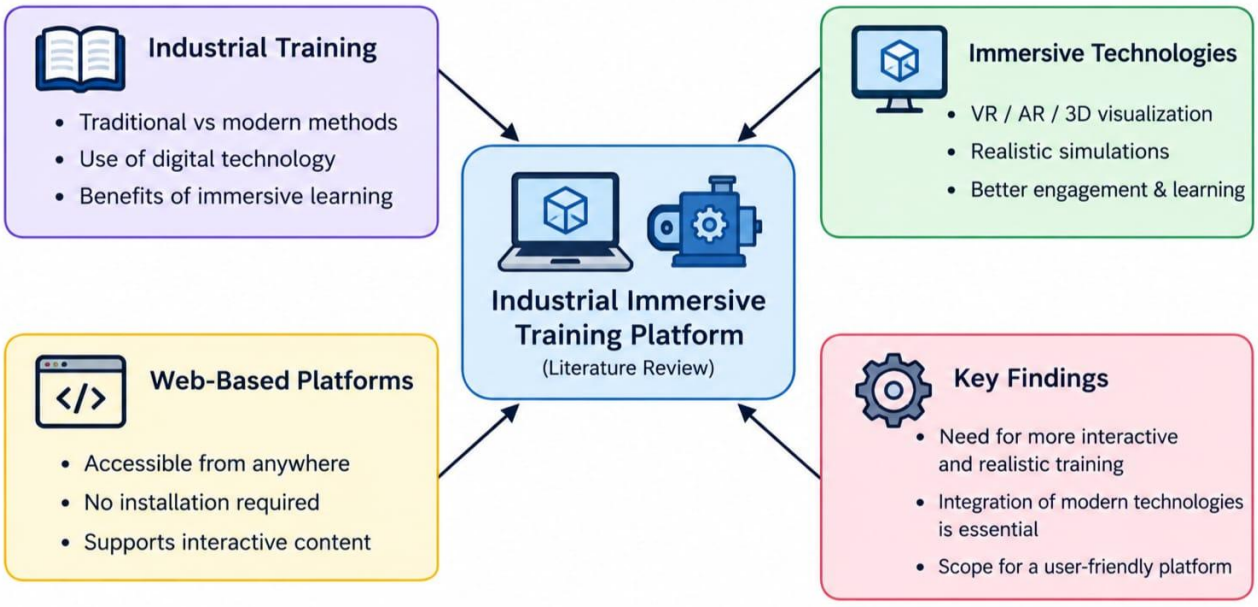


Fig. 2.1: Summary of Literature Review

Table 2.1: Literature Review

Sr. No.	Approach Type (Paper Title / Authors & Year)	Technology Used	Dataset	Key Objective	Limitations
1.	VR Training for KUKA Robot Operators (Pratticò & Lamberti, 2021)	VR, Robotics	Case Study	Train robot operators virtually	Limited real-world testing
2.	AR Tools for Aeronautic Maintenance (Loizeau et al., 2021	AR	Field Evaluation	Support maintenance workers	Industry-specific
3.	Situated Visualization for Maintenance Training (BagheriFard et al., 2025)	AR, Visualization	User Study	Improve maintenance learning	Limited evaluation
4.	Immersive Technologies for Safety Training (Babalola et al., 2023)	VR, AR, XR	67 Articles	Review safety training methods	Varied study quality
5.	XR and Workers' Safety (Dodoo et al., 2025)	XR, VR, AR	Literature Review	Study XR in high-risk industries	Study XR in high-risk industries
6.	VR-ManuSafe (Fengjin et al., 2026)	VR, Unity	8 Injury Types	Teach manufacturing safety	Limited scope

7.	An Emerging Immersive Technology – A Survey (Pavithra et al., 2020)	VR, AR, MR	Literature Survey	Explore immersive technologies	General overview
8.	Immersive Technology: Uses and Challenges (Handa et al.)	VR, AR, MR	Literature Review	Discuss uses and challenges	Limited practical analysis
9.	State of Immersive Technology Research (Suh & Prophet, 2018)	VR, AR, MR	Research Literature	Analyze research trends	Depends on available studies
10.	Web-Based Digital Twin for E-Learning (Weis et al., 2025)	Web 3D, Digital Twin	7 Professionals	Support interactive learning	Small pilot study
11.	Immersive Learning for Machine Operators (Pletz & Zinn, 2020)	VR, Video Analysis	13 Participants	Evaluate operator training	Small sample size
12.	Industrial Machine Identification Using AR (Kavya et al., 2021)	Unity, Vuforia, ARCore	Prototype	Identify machines using AR	Platform limitations

Chapter 3
Information/Chapters

3.1 Introduction to Industrial Machines

Industrial machines are used in manufacturing, processing, assembly, packaging, and other industrial activities. These machines contain different components that work together to perform specific tasks. Understanding their structure and working principles is important for students and industrial trainees.

The proposed Industrial Immersive Training Platform will provide information about selected industrial machines, including their names, uses, components, and basic working principles. This information will help users develop a basic understanding of industrial equipment.



Fig. 3.1: 3D Model of a CNC Milling Machine with Major Components.

3.2 Types of Industrial Machines

Industrial machines can be classified according to their purpose and function. Some common categories are:

1. Manufacturing Machines: Used to shape, cut, or process materials. Examples include drilling machines, milling machines, and lathes.
2. Material Handling Machines: Used to move materials from one location to another. Examples include conveyor systems and industrial lifting equipment.
3. Assembly Machines: Used to join or assemble different components during manufacturing.
4. Packaging Machines: Used to pack products for storage, transportation, and distribution.
5. Industrial Robots: Programmable machines used for tasks such as material handling, welding, and assembly.

The platform may organize machine information into categories so that users can easily find the equipment they want to study.

Table 3.1: Types of Industrial Machines

Sr. No	Type of Machine	Purpose
1	Manufacturing Machine	Used to manufacture products
2	Material Handling Machines	Used to move materials
3	Assembly Machines	Used to join components
4	Packaging Machines	Used to pack products
5	Industrial Robots	Used to perform automated tasks

3.3 Interactive 3D Machine Visualization

Three-dimensional visualization allows users to explore a digital representation of an industrial machine. Unlike a static image, an interactive 3D model can allow users to rotate the model, zoom in and out, and view the machine from different angles.

The proposed platform will aim to provide an interactive 3D viewer through which users can explore supported machine models. Selecting a component may display its name and basic information, depending on the features implemented in the application.

This feature is intended to help users understand the structure and arrangement of machine components in a digital learning environment.

3.4 Machine Component Identification

Industrial machines consist of components that perform different functions. Learning the names and purposes of these components helps students understand the overall machine structure.

The proposed platform may provide component identification activities using labelled 3D models, diagrams, and descriptions. Users will be able to study the available components and learn their basic functions.

For example, a drilling machine may include a drill bit, spindle, worktable, base, and column. The platform can explain the purpose of these components through text and interactive visual elements.

3.5 Simulated Troubleshooting

Troubleshooting is the process of identifying possible causes of a problem and determining suitable corrective actions. It is an important skill in industrial maintenance.

The proposed platform will aim to include simulated troubleshooting exercises based on predefined learning scenarios. These scenarios may describe fictional machine problems and ask users to select an appropriate response from the available options.

This feature will support basic problem-solving practice in a digital environment. The simulations will be educational and will not diagnose actual machine faults or replace qualified maintenance procedures.

3.6 Industrial Safety Training

Industrial safety is essential when working with machines and equipment. Users need to understand safety signs, protective equipment, safe operating practices, and the importance of following approved procedures.

The proposed platform may include safety instructions, multiple-choice questions, and short quizzes related to machine safety. Users can answer questions and review their results to identify topics that need further study.

These activities will support safety awareness but will not constitute official safety certification or replace practical safety training.

3.7 User Management and Training Progress

The platform is planned to support different user roles, such as students, teachers, factory employees, and administrators. User authentication can help provide access to appropriate features according to assigned permissions.

The system may also record quiz scores, completed learning activities, and training progress. A dashboard can present this information so that users can review their learning activities.

Administrators may be provided with features to manage user accounts and machine information, depending on the final implementation.

3.8 Proposed System Workflow

The proposed platform will follow a simple workflow:

1. The user opens the platform through a web browser.
2. The user registers or logs in, if authentication is enabled.

3. The user searches for a machine or selects a machine category.
4. The platform displays the available machine information.
5. The user explores the 3D model, if available.
6. The user completes component identification activities or simulated troubleshooting exercises.
7. The user attempts a safety quiz.
8. The system displays the quiz result and records progress, if this functionality is implemented.

This workflow is intended to make the learning process organized and easy to follow.

3.9 Tools and Technologies

The proposed platform may use the following technologies for development.

Table 3.2: Technologies Used and Their Purpose

Technology	**Purpose**
HTML	Structure of web pages
CSS	Styling and layout of the interface
JavaScript	Application logic and interactivity
React	Development of reusable user interface components
Three.js	Rendering and interaction with 3D models
MySQL Database	To store user and machine data
Visual Studio Code or Antigravity	Writing and managing project code

The final selection of technologies will depend on the project requirements and implementation.

3.10 Applications and Limitations

The Industrial Immersive Training Platform may be useful in diploma colleges, technical institutes, industrial training centres, and organizations that provide introductory machine training.

It can help learners access machine information, explore digital models, and practise basic concepts through quizzes and simulated scenarios. A browser-based design can also reduce the need for specialized VR hardware.

However, the usefulness of the platform will depend on the availability and accuracy of machine models and learning materials. It cannot fully reproduce real machine operation, physical sensations, or every workplace condition. Therefore, it should be used as a supplementary learning tool alongside practical training and approved safety procedures.

Chapter 4
Advantages and Disadvantages

4.1 Advantages

The proposed Industrial Immersive Training Platform offers several potential advantages for students, teachers, and industrial trainees.

1. Interactive Learning: Users can explore digital machine models and learn about machine components in an interactive environment.

2. Easy Accessibility: The platform is planned as a web-based application that can be accessed through a compatible web browser without requiring a dedicated VR headset.

3. Better Understanding of Machines: Machine information, images, and 3D models can help users understand the structure and functions of industrial equipment.

4. Self-Paced Learning: Users can study available learning materials and complete quizzes at their own pace.

5. Safety Awareness: Safety instructions and quizzes can help users learn basic industrial safety practices.

6. Simulated Troubleshooting: Predefined troubleshooting exercises can provide opportunities to practise basic problem-solving in a digital environment.

7. Progress Tracking: Quiz scores and completed activities can help users review their learning progress, if this feature is implemented.

8. Reduced Dependence on Physical Equipment: Digital models can support introductory learning when physical machines are not readily available.

4.2 Disadvantages

Although the proposed platform has several potential benefits, it also has certain limitations.

1. Limited Practical Experience: Digital models cannot fully replace hands-on experience with actual industrial machines.

2. Development Complexity: Creating accurate and interactive 3D machine models may require technical skills and additional development time.

3. Device Compatibility: Some 3D models may require a reasonably capable computer or mobile device for smooth performance.

4. Internet Requirement: Access to online features may depend on a stable internet connection.

5. Limited Simulation: Simulated troubleshooting activities may not represent all real industrial faults or operating conditions.

6. Content Accuracy: Incorrect or incomplete machine information may affect the quality of learning.

7. Maintenance Requirements: The platform may require regular updates to its software, machine models, and learning materials.

8. Safety Limitations: Completing online quizzes does not qualify users to operate industrial machinery independently or replace formal safety training.

4.3 Conclusion

The Industrial Immersive Training Platform aims to make industrial machine learning more interactive and accessible through web-based technology. It may support machine exploration, basic troubleshooting practice, and safety awareness. However, its limitations must be considered, and it should be used as a supplementary learning tool alongside practical workshop training.

Chapter 5
Conclusion

The Industrial Immersive Training Platform is a proposed web-based application designed to support industrial machine learning through interactive digital technology. The platform aims to provide machine information, 3D visualization, component identification, simulated troubleshooting exercises, and safety quizzes in one place.

The main objective of this project is to make industrial training more accessible and interactive for students, teachers, and industrial trainees. By using a web browser, users can explore available machine models and learn about their components without requiring a dedicated Virtual Reality (VR) headset.

The proposed platform may help users develop a basic understanding of industrial machines and improve their awareness of machine components and safety practices. Features such as quizzes and progress tracking can also support self-paced learning.

However, the platform cannot completely replace practical experience with real industrial equipment. Its usefulness will depend on the quality of the 3D models, accuracy of the learning content, and successful implementation of its features.

In conclusion, the Industrial Immersive Training Platform aims to combine web technology and interactive 3D visualization to support modern technical education. In the future, the platform could be expanded with additional machine models, more training activities, and improved interactive features.

Chapter 6
Project Initiation

6.1 Description of Problem Statement

Industrial training is an important part of technical education, but students may not always have access to different industrial machines because of their high cost, limited availability, and maintenance requirements. Learning only through textbooks and static images may also make it difficult to understand machine components and their working principles.

Therefore, there is a need for an interactive learning platform that provides machine information and digital visualization. The proposed Industrial Immersive Training Platform aims to address this problem by providing a web-based environment where users can explore industrial machines, identify components, practise simulated troubleshooting, and learn basic safety procedures.

6.2 Scope and Objectives

Scope:

The scope of the proposed project is to develop a web-based platform for basic industrial machine training. It will aim to provide interactive learning materials and digital machine models that users can access through a compatible web browser.

The proposed platform may include machine categories, machine search, 3D visualization, component identification, simulated troubleshooting exercises, safety quizzes, user accounts, and training progress tracking. It is intended for educational purposes and will supplement practical workshop training.

Objectives:

1. To develop a web-based industrial training platform.
2. To provide information about different industrial machines and their components.
3. To enable users to explore available machine models through interactive 3D visualization.

4. To provide basic component identification activities.
5. To support learning through simulated troubleshooting exercises.
6. To improve safety awareness through educational quizzes.
7. To provide a convenient way to review learning progress, where implemented.

6.3 Stakeholders

The main stakeholders of the proposed Industrial Immersive Training Platform are:

1. Students: Use the platform to learn about industrial machines and their components.
2. Teachers: Use the platform as a supplementary teaching resource.
3. Factory Employees: Use the available learning materials to revise basic machine-related concepts.
4. Institution Administrators: Manage institution-related users and learning resources, if supported.
5. Platform Administrators: Manage user accounts, machine information, and platform content.
6. Project Developers: Design, develop, test, and maintain the application.

6.4 Seminar Discussion Plan

The seminar discussion plan describes the main topics that will be presented during the seminar.

1. Introduction to industrial immersive technology.
2. Need for interactive industrial machine training.
3. Overview of the Industrial Immersive Training Platform.
4. Explanation of the proposed system and its features.
5. Technologies required for developing the platform.
6. Advantages and disadvantages of the proposed system.
7. Possible applications and future improvements.
8. Conclusion and questions from the audience.

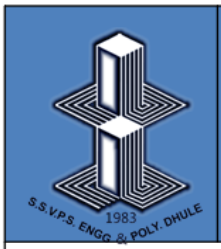
6.5 Platform / Equipment / Resources Identification

The following resources may be required to develop and test the proposed platform.

Table 6.1: Resource Used and Their Purpose

Resource	**Purpose**
Laptop or Computer	To develop and test the application.
Antigravity	To assist with application development and coding.
HTML and CSS	To structure and style web pages.
JavaScript	To implement application logic and interactivity.
React	To build the user interface.
Three.js	To display and interact with 3D machine models.
MySQL Database	To store user and machine data.
3D Machine Models	To provide digital representations of industrial equipment.
Web Browser	To access and test the application.
Internet Connection	To access online tools, libraries, and services when required.

Chapter 7
Seminar Presentation

	**S.S.V.P.S's B. S. Deore Polytechnic, Dhule** **Department of Computer Engineering** *Academic Year : 2026-2027*

SEMINAR AND PROJECT INITIATION COURSE

Sem : CO 5 K	**SPI – 315003**	**Date :**

Topic : Industrial Immersive Training Platform

Roll No.	Name of Student
14	Purva Vikas Bhamare
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39	Prajakta Chetan Mahale
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   - 6.5 Platform/ Equipment/ Resources identification.

1. Bibliography
2. References

Abstract

- Immersive technology is becoming useful in education, industrial training, and technical skill development.
- Traditional training needs physical machines, trained instructors, and workshop facilities, so students get limited practical exposure.
- The Industrial Immersive Training Platform is a web-based application to learn industrial machines through interactive digital content.
- Features: machine search, machine information, interactive 3D visualization, safety instructions, simulated troubleshooting, quizzes, and progress tracking.
- Runs in a web browser, with no VR headset required. Intended for education only; it does not replace workshop training or official safety procedures.

3

1. Introduction

- Industrial training helps students understand machines, their components, working principles, and safety procedures.
- Access to real machines is limited by high cost, maintenance needs, and safety risks.
- VR, AR, and interactive 3D visualization let learners explore digital machine models.
- Proposed project: a web-based application with machine information, 3D models, component identification, simulated troubleshooting, safety quizzes, and progress tracking.
- Objective: make machine learning more accessible and interactive, while supporting (not replacing) workshop training.

4

2. Literature Survey

Sr. No.	Aapproach Type	Technology Used	Dataset	Key Objective	Limitations
1.	VR Training for KUKA Robot Operators (Pratticò & Lamberti, 2021)	VR, Robotics	Case Study	Train robot operators virtually	Limited real-world testing
2.	AR Tools for Aeronautic Maintenance (Loizeau et al., 2021)	AR	Field Evaluation	Support Maintenance Workers	Industry-specific
3.	Situated Visualization for Maintenance Training (BagheriFard et al., 2025)	AR, Visualization	User Study	Improve maintenance learning	Limited evaluation

2. Literature Survey

Sr. No.	Aapproach Type	Technology Used	Dataset	Key Objective	Limitations
4.	Immersive Technologies for Safety Training (Babalola et al., 2023)	VR, AR, XR	67 Articles	Review safety training methods	Varied study quality
5.	XR and Workers' Safety (Dodoo et al., 2025)	XR, VR, AR	Literature Review	Study XR in high-risk industries	Adoption challenges
6.	VR-ManuSafe (Fengjin et al., 2026)	VR, Unity	8 Injury Types	Teach manufacturing safety	Limited scope

2. Literature Survey

Sr. No.	Aapproach Type	Technology Used	Dataset	Key Objective	Limitations
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8.	Immersive Technology: Uses and Challenges (Handa et al.)	VR, AR, MR	Literature Review	Discuss uses and challenges	Limited practical analysis
9.	State of Immersive Technology Research (Suh & Prophet, 2018)	VR, AR, MR	Research Literature	Analyze research trends	Depends on available studies

2. Literature Survey

Sr. No.	Aapproach Type	Technology Used	Dataset	Key Objective	Limitations
10.	Web-Based Digital Twin for E-Learning (Weis et al., 2025)	Web 3D, Digital Twin	7 Professionals	Support interactive learning	Small pilot study
11.	Immersive Learning for Machine Operators (Pletz & Zinn, 2020)	VR, Video Analysis	13 Participants	Evaluate operator training	Small sample size
12.	Industrial Machine Identification Using AR (Kavya et al., 2021)	Unity, Vuforia, ARCore	Prototype	Identify machines using AR	Platform limitations

3. Information to seminar topic

- **Industrial machines:** used in manufacturing, processing, assembly, and packaging; the platform gives each machine's name, use, components, and working principle.
- **Types:** Manufacturing (drilling, milling, lathe), Material Handling (conveyors, lifting), Assembly, Packaging, and Industrial Robots.
- **3D visualization:** rotate, zoom, and view a machine from different angles; selecting a component shows its name and basic information.
- **Component identification:** labelled 3D models and diagrams, e.g. a drilling machine has a drill bit, spindle, worktable, base, and column.





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4. Advantages / Disadvantages

Advantages

- Interactive learning
- Easy access through a browser, no VR headset
- Better understanding of machine structure
- Self-paced learning
- Safety awareness
- Simulated troubleshooting practice
- Progress tracking
- Less dependence on physical equipment

Disadvantages

- Cannot replace hands-on experience
- 3D models take skill and time to build
- Needs a reasonably capable device
- Needs a stable internet connection
- Limited simulation of real faults
- Content accuracy affects learning
- Needs regular updates
- Quizzes do not replace formal safety training

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5. Conclusion

- The Industrial Immersive Training Platform is a proposed web-based application that supports industrial machine learning through interactive digital technology.
- It brings machine information, 3D visualization, component identification, simulated troubleshooting, and safety quizzes into one place.
- Users can explore machine models and learn about components in a web browser, without a VR headset.
- It cannot fully replace real equipment experience; its value depends on model quality, content accuracy, and successful implementation.
- Future scope: more machine models, more training activities, and improved interactive features.

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6. Project Initiation

- **6.1** Description of problem statement
- **6.2** Scope and objectives
- **6.3** Stakeholders
- **6.4** Seminar discussion plan
- **6.5** Platform / equipment / resources identification

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6.1 Description of problem statement

- Industrial training is important in technical education, but students may not get access to different machines because of high cost, limited availability, and maintenance needs.
- Learning only from textbooks and static images makes machine components and working principles hard to understand.
- **Need:** an interactive learning platform with machine information and digital visualization.
- **Solution:** a web-based environment to explore machines, identify components, practise simulated troubleshooting, and learn basic safety procedures.

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6.2 Scope and objectives

Scope

- A web-based platform for basic industrial machine training with machine categories and search, 3D visualization, component identification, simulated troubleshooting, safety quizzes, user accounts, and progress tracking. For education only; supplements workshop training.

Objectives

- Develop a web-based industrial training platform
- Provide information on machines and their components
- Enable interactive 3D exploration of machine models
- Provide basic component identification activities
- Support learning through simulated troubleshooting
- Improve safety awareness through quizzes
- Let users review their learning progress

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6.3 Stakeholders

- **Students:** learn about industrial machines and their components.
- **Teachers:** use the platform as a supplementary teaching resource.
- **Factory employees:** revise basic machine-related concepts.
- **Institution administrators:** manage institution users and learning resources.
- **Platform administrators:** manage user accounts, machine information, and content.
- **Project developers:** design, develop, test, and maintain the application.

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6.4 Seminar discussion plan

- Introduction to industrial immersive technology
- Need for interactive industrial machine training
- Overview of the Industrial Immersive Training Platform
- Proposed system and its features
- Technologies required for development
- Advantages and disadvantages
- Possible applications and future improvements
- Conclusion and questions from the audience

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6.5 Platform / Equipment / Resources

- **Laptop / Computer:** develop and test the application
- **Antigravity:** assist with application development and coding
- **HTML and CSS:** structure and style web pages
- **JavaScript:** application logic and interactivity
- **React:** build the user interface
- **Three.js:** display and interact with 3D machine models
- **MySQL Database:** store user and machine data
- **3D Machine Models:** digital representations of industrial equipment
- **Web Browser:** access and test the application
- **Internet Connection:** online tools, libraries, and services

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SEMINAR and PROJECT INITIATION Assessment / Evaluation

Sr. No.	Enrollment Number	Seat Number	Name of student
1	24610960181	388501	Purva Vikas Bhamare
2	24610960213	388521	Shubhangi Sanjay Kedar
3	24610960218	388526	Prajakta Chetan Mahale
4	24610960225	388533	Mansi Pradip Nandre

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388501	388521	388526	388533
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign:

Name: **Shraddha S. Hajari**

(Course Expert/s)